Project Iteration 01 Design Team Trainer App

Analyze and Apply Principles of Computing to design Solutions using concepts from CS1050 class lectures following CS1050 Agile Software Development Lecture

Points: 20pt (see rubric in canvas)

Due Date: See Canvas.
Work will be accepted up to 24 hours after the due date with a 10% penalty. Meaning if you turn it in at 12:01 am of the next day you will be deducted 10% of the total points from your score.
If the assignment is more than 24 hours late, it will be a 0.

Effort: Individual (Academic Integrity) .

- **Write your own design and answer the questions in your own words** . **Do not use AI for this assignment.** Handing in something that is not your own work is not beneficial to your learning and will be a violation.

Deliverables: Upload this document with your answers and your professional design and testing plan in a separate document.

Project Team and Athlete Analysis

Description: This program asks the trainer for the number of athletes, then the trainer enters each athlete's name, weight, height, and age. The program calculates and stores each athlete's BMI and max heart rate (MHR), displays each athlete's results, calculates the average max heart rate, lists athletes outside the normal BMI range, lists athletes above or equal to the average max heart rate, and optionally calculates training heart rates for the team.

Requirements: User Stories and Acceptance Criteria

User story 1: View Program Overview

As a trainer, I want to see an overview of what the program does and the BMI category ranges so that I understand how BMI will be interpreted.

Display the following:

```
*************************************
Program Overview
*************************************
The trainer enters how many athletes are on the team.
Then the trainer enters each athlete's weight, height and age.
For each athlete, the program calculates BMI and Max Heart Rate.
BMI Categories
Under 18.5: Underweight
18.5 to under 30: Normal
30 or greater: High

Calculates percentage of max heart rate for athlete training goal if needed
```

User Story 2: Specify Number of Athletes

As a trainer, I want to enter the number of athletes before starting so that the program repeats the data entry process for each athlete.

Acceptance Criteria:

- Prompts for number of athletes
- Input must be valid positive integer
- Displays error and repeats question if invalid number of athletes entered

```
Enter the number of athletes on the team: 0
Error: value must be greater than 0.
Enter the number of athletes on the team: 3
```

User Story 3: Validate Athlete age, height and weight

As a trainer, I need a fitness tracking system to enter each athlete's fitness data (weight, height, age) so the program can correctly calculate and analyze BMI and Max Heart Rate.

- Name should be first name only and no spaces
- Weight and height can contain decimal values
- Age will not contain decimal values
- Age, height and weight must be > 0
- Displays error and repeats until valid
- Do not need to store all the athlete's ages, heights and weights values for later. You are using the entries to to store a list of the athlete's BMI and Max Heart Rate (MHR)

```
Enter data for athlete 1
Enter athlete's first name: Gordon
Enter weight in pounds: -1
Error: value must be greater than 0.
Enter weight in pounds: 235
Enter height in inches: 80
Enter age in years: 0
Error: value must be greater than 0.
Enter age in years: 30

Enter data for athlete 2
Enter athlete's name: Jokic
Enter weight in pounds: 284
Enter height in inches: 83
Enter age in years: 30
```

User Story 4: Calculate and Store BMI

As a trainer, I want the program to calculate each athlete's BMI and store it so that the program can analyze the team results.

- BMI formula used: $703 \times \text{weight} / \text{height}^2$
- Each athlete's BMIs are stored for analysis after all BMIs calculated

User Story 5: Calculate and Store Max Heart Rate

As a trainer, I want the program to calculate each athlete's max heart rate and store it so that the program can analyze the team results.

- Max Heart Rate: 220 -age
- Max Heart Rates stored for analysis after all Max Heart Rates calculated
- Repeat user stories 2 through 5 until all athletes have been entered.

User Story 6: Display Athlete BMI Value, Category and Max Heart Rate

As a trainer, I want the program to display each athlete's BMI value, BMI category and Max Heart Rate so that I can review individual results.
Acceptance Criteria:

- Displays the athlete name, BMI formatted to 1 decimal place and MHR.

```
========== Team Summary==========

Gordon
BMI: 25.8
Category: Normal
MHR: 190

Jokic
BMI: 29.0
Category: Normal
MHR: 190

Peyton
BMI: 22.0
Category: Normal
MHR: 197

Bruce
BMI: 24.6
Category: Normal
MHR: 192

Deb
BMI: 43.1
Category: High
MHR: 161

Monty
BMI: 17.6
Category: Underweight
MHR: 200
```

User Story 7: List all athletes outside normal BMI range

As a trainer, I want the program to list athletes that are above or below the BMI category of normal so I can design different training plans.
Acceptance Criteria:

- Displays Athlete names and whether above or below normal

```
Above normal: Deb
Below normal: Monty
```

- When none outside range display “No athletes outside of normal range”

```
No athletes outside of normal range
```

User Story 8: Calculate and Display Average of the Max Heart Rates

As a trainer, I want the program to calculate and display the average BMI so that I can understand the overall group trend.
Acceptance Criteria:

- Calculates average Max Heart Rates of the team.
- Displays average formatted to 1 decimal place

```
Average Max Heart Rates: 188.3
```

User Story 9: Identify Athlete with Highest Max Heart Rate

As a trainer, I want the program to determine which athlete has the highest max heart rate so that I can identify the athlete with the highest cardiovascular capacity estimate.

Acceptance Criteria

- You may assume that only one athlete will have the highest max heart rate.
- You do not need to handle duplicate highest values.
- The program determines the index of the athlete with the highest max heart rate.
- The program uses the index to display:
  - athlete name
  - max heart rate

```
Monty has highest max heart rate: 200
```

User Story 10: List all athletes above or equal to Max Heart Rate average

As a trainer, I want the program to list athletes that are above or equal to the max heart rate average so I can design a different training plan.
Acceptance Criteria:

- Display athletes above or equal to average MHR

```
Athletes above or equal to average MHR:
Gordon
Jokic
Peyton
Bruce
Monty
```

User Story 11: Calculate and Display Training Heart Rate

As a trainer, if needed I want to enter one training percentage for the team so that the program can calculate and display each athlete's training heart rate.

Acceptance Criteria:

- prompts trainer if they want to calculate a training heart rate for all athletes and checks that valid data has been entered: 'y', 'Y', 'N' or 'n'
    - When not a y or n character repeat question
- 'y' or 'Y' Ask for training percentage and validate value
    - percentage must be greater than 0
    - displays error and repeats if invalid
    - If valid calculates and displays name training heart rate to 1 decimal place
- 'n' or 'N' Display Training Program Analysis complete

```
Do you want to calculate the training heart rates? (y/n): g
Error: enter y, Y, n, or N.
Y

Enter training percentage: 0
Error: value must be greater than 0.
Enter training percentage: 85

Gordon Training Heart Rate: 161.5
Jokic Training Heart Rate: 161.5
Peyton Training Heart Rate: 167.5
Bruce Training Heart Rate: 163.2
Deb Training Heart Rate: 136.9
Monty Training Heart Rate: 170.0
```

Training Program Analysis complete
Do you want to calculate the training heart rates? (y/n): no Training Program Analysis complete

Example Output

```
*************************************
Program Overview
*************************************
The trainer enters how many athletes are on the team.
Then the trainer enters each athlete's weight, height and age.
For each athlete, the program calculates BMI and Max Heart Rate.
BMI Categories
Under 18.5: Underweight
18.5 to under 30: Normal
30 or greater: Overweight
Calculates percentage of max heart rate for training

*************************************
Athlete Entry
*************************************
Enter the number of athletes on the team: 6

Enter data for athlete 1
Enter athlete's name: Gordon
Enter weight in pounds: 235
Enter height in inches: 80
Enter age in years: 30

Enter data for athlete 2
Enter athlete's name:Jokic
Enter weight in pounds: 284
Enter height in inches: 83
Enter age in years: 30

Enter data for athlete 3
Enter athlete's name: Peyton
Enter weight in pounds: 200
Enter height in inches: 80
Enter age in years: 23

Enter data for athlete 4
Enter athlete's name: Bruce
```

```
Enter weight in pounds: 202
Enter height in inches: 76
Enter age in years: 28

Enter data for athlete 5
Enter athlete's name: Deb
Enter weight in pounds: 275
Enter height in inches: 67
Enter age in years: 59

Enter data for athlete 6
Enter athlete's name: Monty
Enter weight in pounds: 130
Enter height in inches: 72
Enter age in years: 20

========== Athlete Summary ==========
Gordon
BMI: 25.8
Category: Normal
MHR: 190

Jokic
BMI: 29.0
Category: Normal
MHR: 190

Peyton
BMI: 22.0
Category: Normal
MHR: 197

Bruce
BMI: 24.6
Category: Normal
MHR: 192

Deb
BMI: 43.1
Category: High
MHR: 161

Monty
BMI: 17.6
Category: Underweight
MHR: 200

========== BMI Analysis ==========
BMI Above normal: Deb
BMI Below normal :Monty
```

```
========== MHR Analysis ==========
Team Average Max Heart Rates: 188.3

Monty has highest max heart rate: 200

Athletes above or equal to average MHR:
Gordon
Jokic
Peyton
Bruce
Monty

Do you want to calculate the training heart rates? (y/n): g
Error: enter y, Y, n, or N.
y

Enter training percentage: 85

Gordon Training Heart Rate: 161.5
Jokic Training Heart Rate: 161.5
Peyton Training Heart Rate: 167.5
Bruce Training Heart Rate: 163.2
Deb Training Heart Rate: 136.9
Monty Training Heart Rate: 170.0

**************************************
Training Program Analysis Complete
**************************************
```

2 Software Development

Review [Agile Software Development CS1050]

You are not being graded that your algorithms will all work but on how you followed the software development process to come up with test cases and algorithms. After this assignment you will collaborate in class on improving your design before implementing your code.

- Analyze user story and acceptance criteria requirements and break the problem into smaller tasks and subtasks
- Create Varied Unit Test Cases to ensure acceptance criteria
- Use test cases design method algorithms in pseudocode for each task

When you are asked to draw a picture, draw it on paper, whiteboard or tablet (not typed) Crop pictures so they are readable.

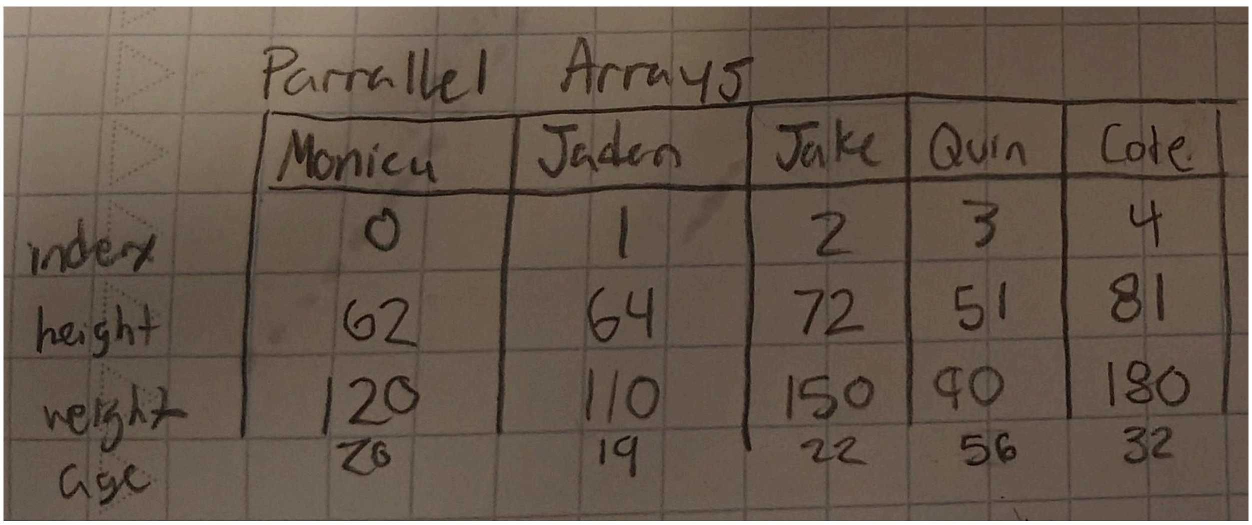
YOU ARE NOT WRITING ANY CODE - YOU ARE ONLY COMPLETING THE DESIGN.

2.1 Analyze User Stories and Acceptance Criteria

Answer the following based on what you started in class.

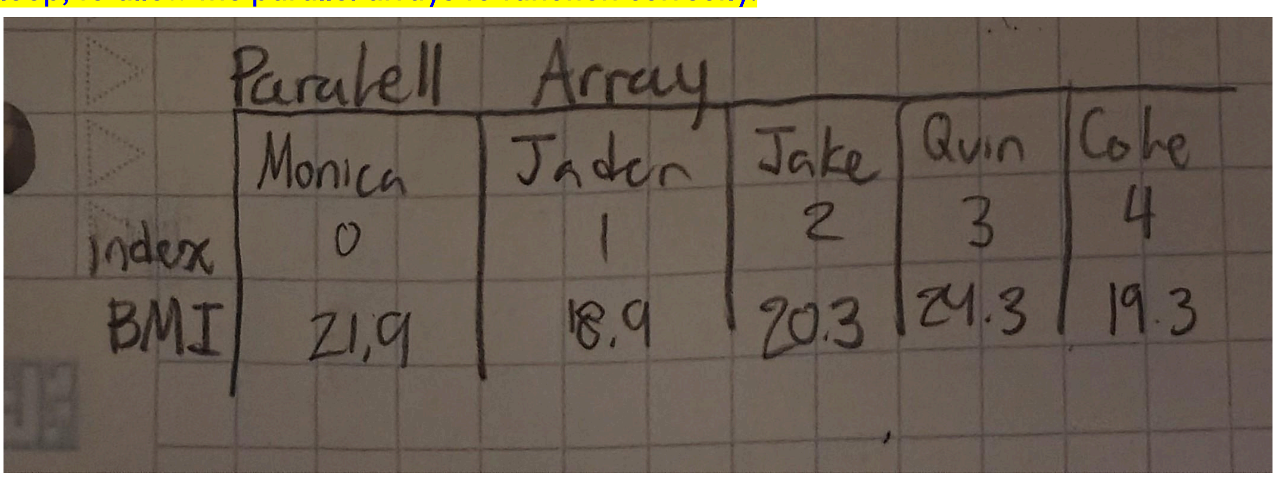
1. What will you use to store each athlete's weight and height? Include drawing.

Parallel Arrays because they would allow me to set each Athlete's stats to one index in the arrays.



2. What will you use to store all BMI values? Include drawing.

Parallel Arrays because they would allow me to set BMI to an index number that corresponds with their height, weight, and name. User stories 1-5 would all be put in a for loop, to allow the parallel arrays to function correctly.



3. What constants will you need?

Q4 down below

4. Draw the flow of major tasks and subtasks.

Q3 down below

5. List the variables and arrays needed in main.

Q4 down below.

6. List the methods you need (return type, name, parameters).

Weight, height, age, and bmi are all double arrays.

1. What data must be stored about each athlete in an array? What data types? Weight, height, age, and bmi are all double arrays. String arrays are category and names. Int max heart rate. What other values must be stored in main? What data types? The number of athletes is stored in main and it is an int. Average MHR is double.
2. What is SRP and DRY? How do you design a modular solution that is easier to maintain and test? SRP is the different methods making it easier to test each one individually to make sure it works. DRY is the checkAboveZero which checks the input values to make sure they're not negative or 0.
3. Look at user stories and the acceptance criteria to break the problem into methods to do the tasks. Are there any similar tasks that can be implemented with one method? Checking for a valid input is a task that can be used multiple times. It is referenced enough that it should have its own method. double checkAboveZero(double value) List the tasks that you think you will be creating methods for. Include a brief description, methods name, parameters and return value.

void programOverview()

int numberOfAthletes(Scanner athleteInput)

double[] calcBMI(double[] heightForBMI, double[] weightForBMI)

double[] calcMHR(double[] ageForMHR)

String[] determineAndDisplayCategory(double[] bmiForCategory)

void outNormalBMI(double[] bmiNoNorm, String[] nameNoNorm)

double averageMHR(double[] mhrGiven)

int highestMHR(double[] mhrPassed)

void displayGreaterEqMHR(double averageMHRGiven, String[] nameAvg)

void diplayCalcTMHR(double[] mhrCalc, String[] namesTMHR)

I forgot to add another method with the signature: double checkAboveZero(double value)

1. Declare variables / Arrays in main

2. Program Overview
   - Summarize what the program does

3. Input Athlete stats.
   - Initialize # of Athletes
   - Allocate array memory for:
     - Names
     - Height
     - Weight
     - Age
     - BMI
     - Category
     - MHR
   - For loop for Names, Heights, weight, Age

4. Calculate BMI
   - Pass height, weight arrays
   - (703 * weight)/ height^2
   - return double array

6. BMI category
   - Pass BMI[]
   - If statement for category
   - return String[] category

5. MHR Calc
   - Pass age array
   - 220 - age (for loop)
   - return double[] MHR

7. Non-Normal BMI Category
   - if BMI cat ≠ normal then if BMI < 18.5 and if BMI > 24.9
     Below Above
   - if not then "No athletes"

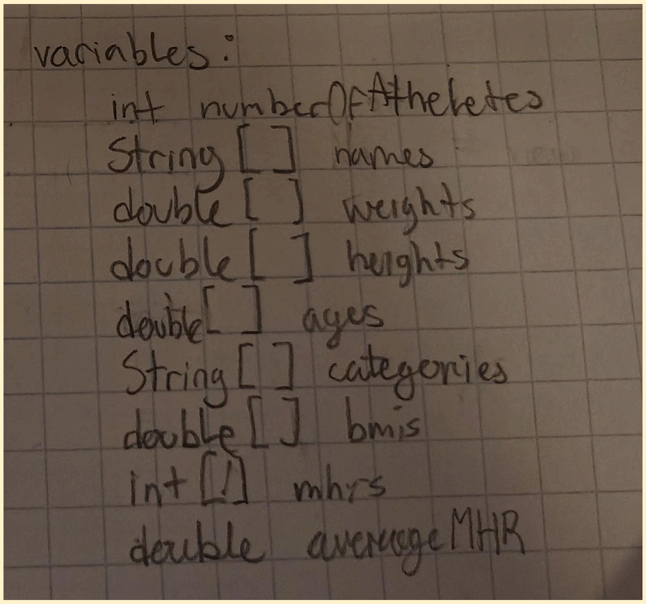
8. Calc Avg MHR
   - pass MHR[]
   - sum MHR / MHR.length
   - Return double

9. Highest MHR
   - pass MHR[] & names[] for display
   - highest < MHR[i] highest = MHR[i]
   - Print highest.

10. Above Average MHR
   - if MHR[i] > MHRavg
     print "Name"

11. Training Heart Rates
   - Y or N prompt if Y continue, if not message → end
   - Percentage input
   - Names and MHR
   - Print "Name

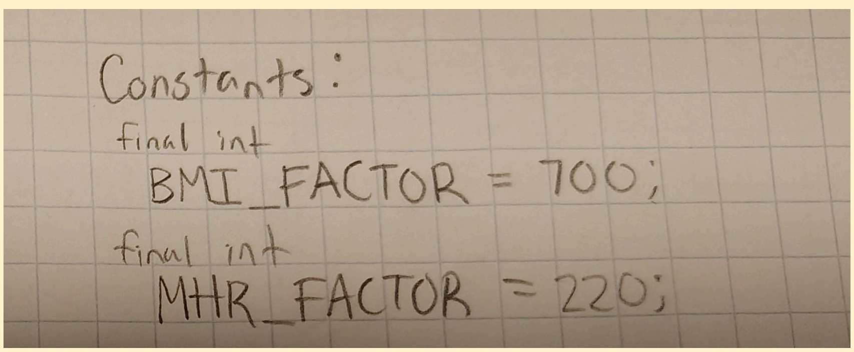
4. What is the flow of main to do these tasks? Write the declarations below for all the variables and constants, including array variables. Create pseudocode in main on paper or white board showing the flow using the variables and methods you identified above.



variables:
int numberOfAthletes
String [ ] names
double [ ] weights
double [ ] heights
double [ ] ages
String [ ] categories
double [ ] bmis
int [ ] mhrs
double averageMHR

Constants needed are the numbers set for the formulas for both BMI and MHR.

I MEANT 703 HERE



Constants:
final int
BMI_FACTOR = 700;
final int
MHR_FACTOR = 220;

2.2 Create Test Cases, Methods and Algorithms in Pseudocode

Create a new document for your design and testing plan. This is a professional communication outlining your plan. You can copy and paste information from other assignments. During the lectures you will be given more direction on what should be in your design.

Design Unit Tests and Methods :
- Unit tests: table containing Pre/Post conditions can be handwritten or in document
- Method signature return methodName(list parameters)

- Handwritten pseudocode for method algorithms on paper, white board or tablet for pseudocode to get credit. You can use an image of your group design from the presentation but not other groups.
- Remember to include your design for the main method you did in part 2.1

You can copy and paste the following template to organize your plan.

User Story: User Story 8: Calculate and Display Average of the Max Heart Rates

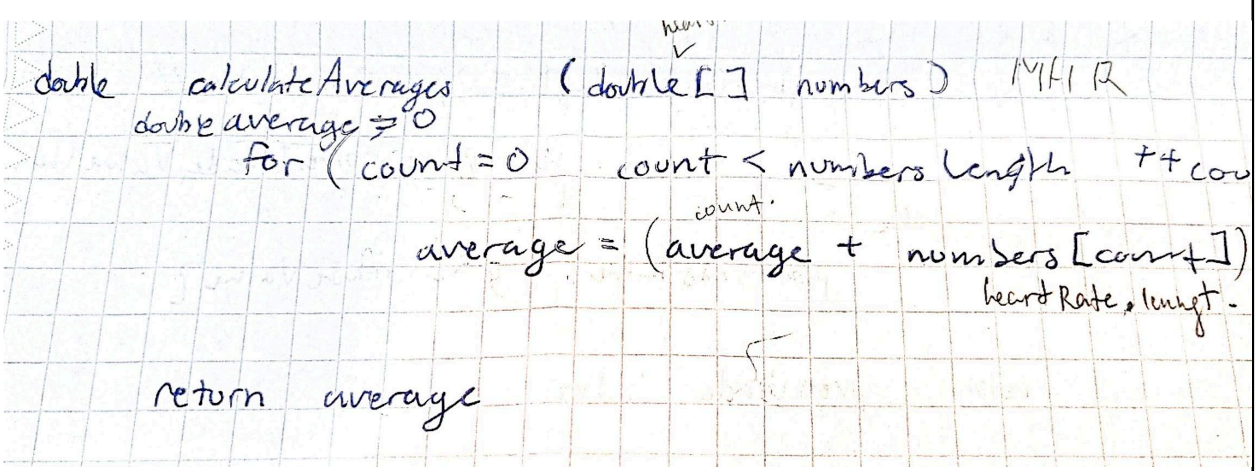
Method Description:

- Calculates average Max Heart Rates of the team.
- Displays average formatted to 1 decimal place

Pre	Post
MHR Array : [180, 120, 190, 10]	Average = 125
MHR Array : [20]	Average = 20
MHR Array : [180, 150, 120, 100, 110, 170]	Average = 138.333333
MHR Array : [300, 300]	Average = 300

Method Signature: double calculateAverage (double[] MHR)

Design: Put an image of your pseudocode handwritten on paper, white board or tablet for the algorithm.



1. What part of the design was the most difficult? Explain why.

The most difficult part was most definitely the for loop. Trying to find the right range for the array was difficult.
2. What part of the design do you feel most confident about? Explain why. I feel the most confident about our parameters and return value. This is because they are both very basic
3. How did your group collaboration help you with your design? They helped develop the method Signature and asked some very important questions about for loops. They noticed a couple of bugs that I missed, specifically with the starting value of count in the for loop.